ELEC 221 Lecture 11
The CT Fourier Transform

Thursday 15 October 2026

Announcements

- Practice midterms available on PrairieLearn by end-of-week
- Quiz 6 Tuesday
- Tutorial assignment 3 on Monday (image processing)

Where are we going?

Today

Learning outcomes:

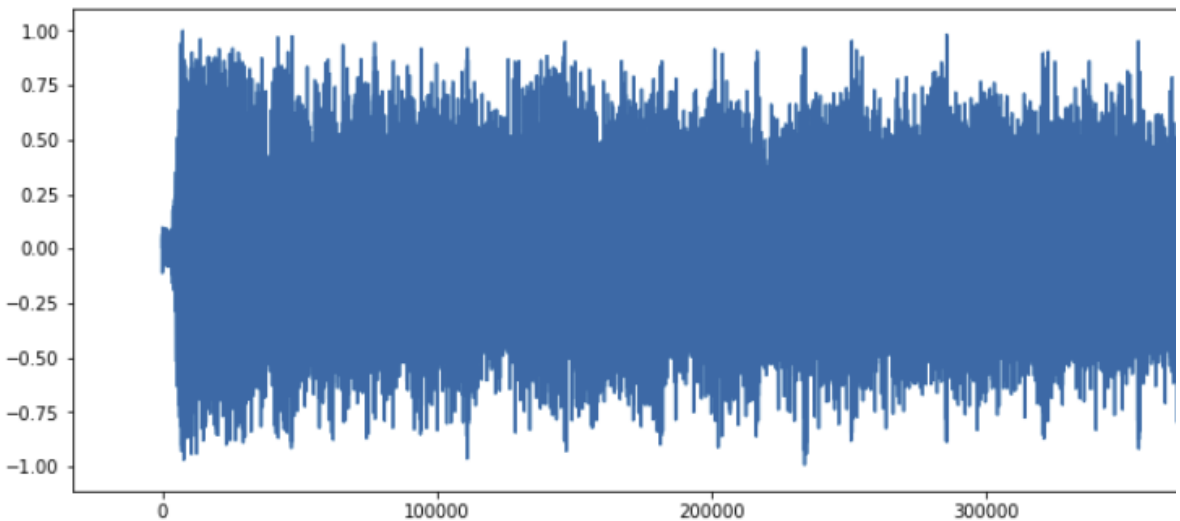
- Distinguish between the CT Fourier series and Fourier transform
- Define the CT Fourier transform and compute the Fourier spectrum of a CT signal
- State sufficient criteria for a signal to have a Fourier transform

Fourier series

In tutorial assignment 2, you worked with signals that *weren't periodic*:

```
plt.plot(audio)
```

```
[<matplotlib.lines.Line2D at 0x7f940880e370>]
```



But, we were still doing *something* with Fourier analysis:

```
fourier_spectrum = np.fft.rfft(audio)
```

The Fourier transform

The **Fourier transform** generalizes the Fourier series to **aperiodic signals**. It involves all possible frequencies.

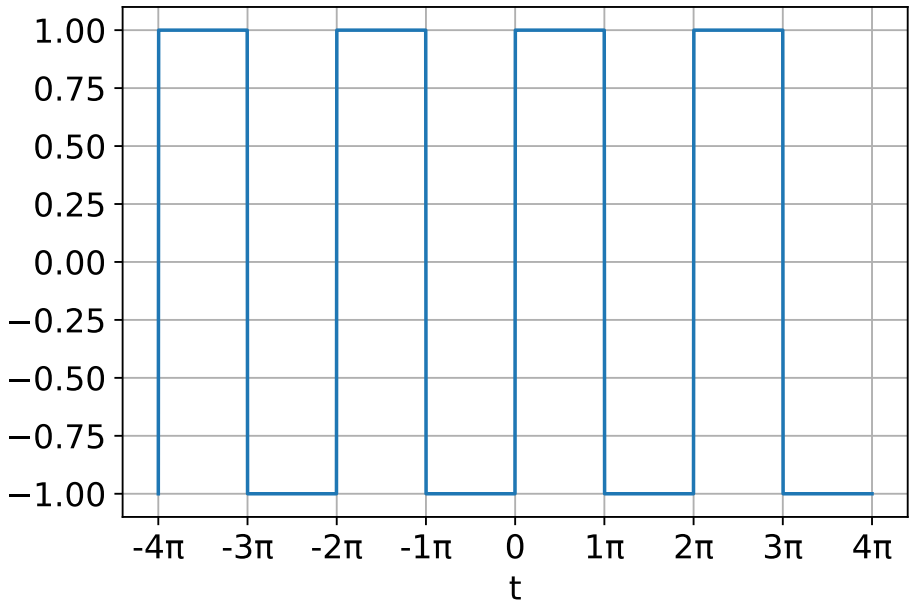
Fourier series:

Fourier transform:

How do we get here?

Towards the Fourier transform

Previously analyzed a 2π-periodic square wave:



We derived its Fourier series representation

Towards the Fourier transform

Let's generalize this. Consider the following square wave:

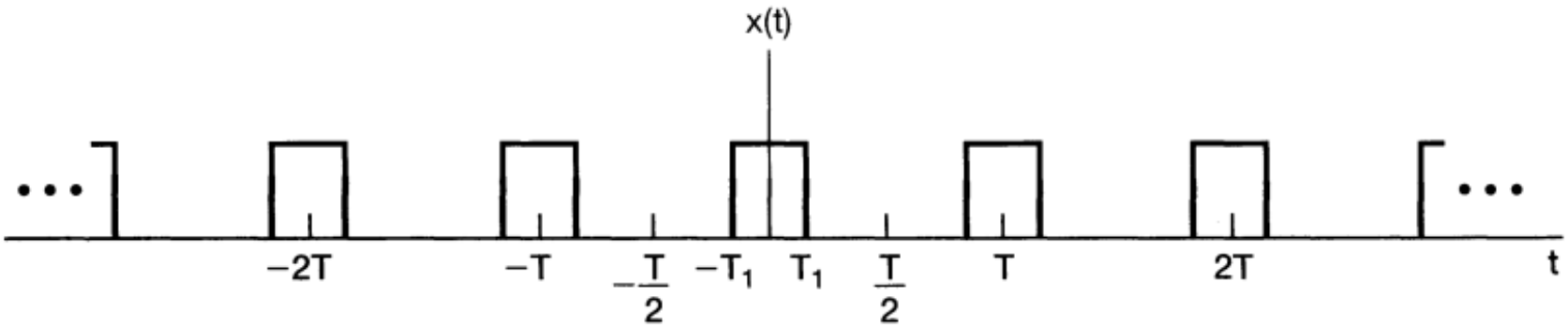


Image credit: Oppenheim chapter 4.1

Towards the Fourier transform

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases}$$

Compute its Fourier coefficients.

$$c_k = \frac{1}{T} \int_T x(t) e^{-jk\omega t}$$

Start with c_0:

Towards the Fourier transform

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < T/2 \end{cases}$$

Now the c_k:

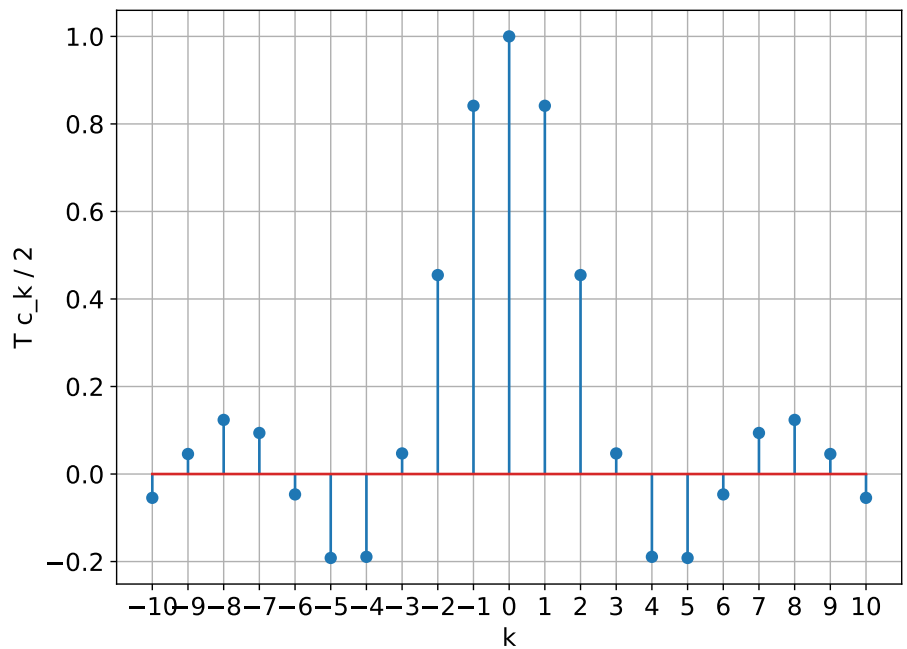
Towards the Fourier transform

What does this function look like?

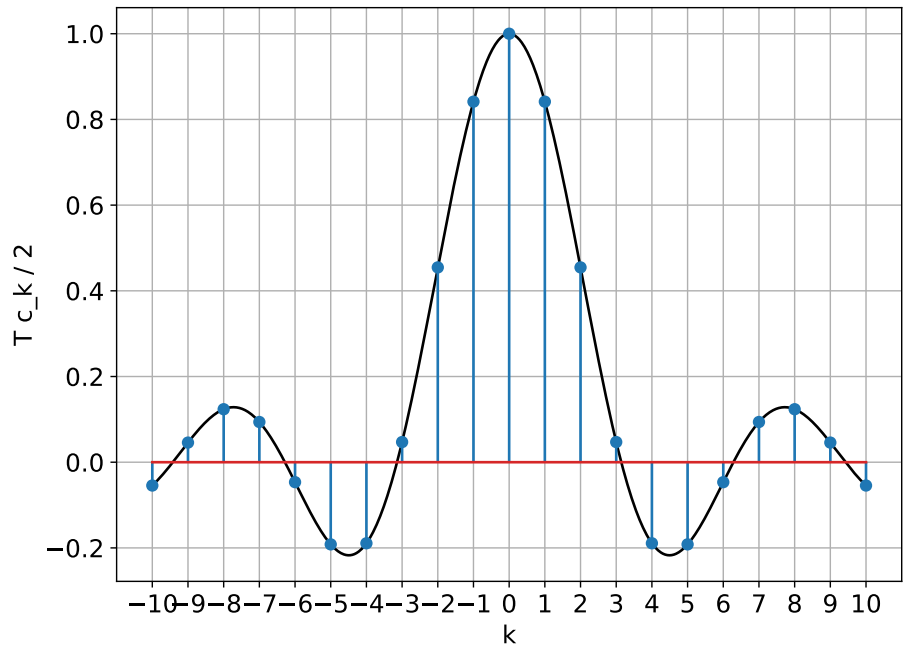
Rearrange and express as a function of k:

Let's plot this: set $T_1 = 1$, and $T = 2\pi$, so $\omega = \frac{2\pi}{T} = 1$.

Towards the Fourier transform



Towards the Fourier transform



These are **samples** of the function

at *integer values of* k.

Towards the Fourier transform

Re-express this as a function of frequency, $\tilde{\omega} = k\omega$:

The Fourier coefficients are samples of this function at *integer multiples* of fundamental frequency, $\tilde{\omega} = k\omega$, where $\omega = 2\pi/T$.

Towards the Fourier transform

Suppose T grows, but T_1 stays the same:

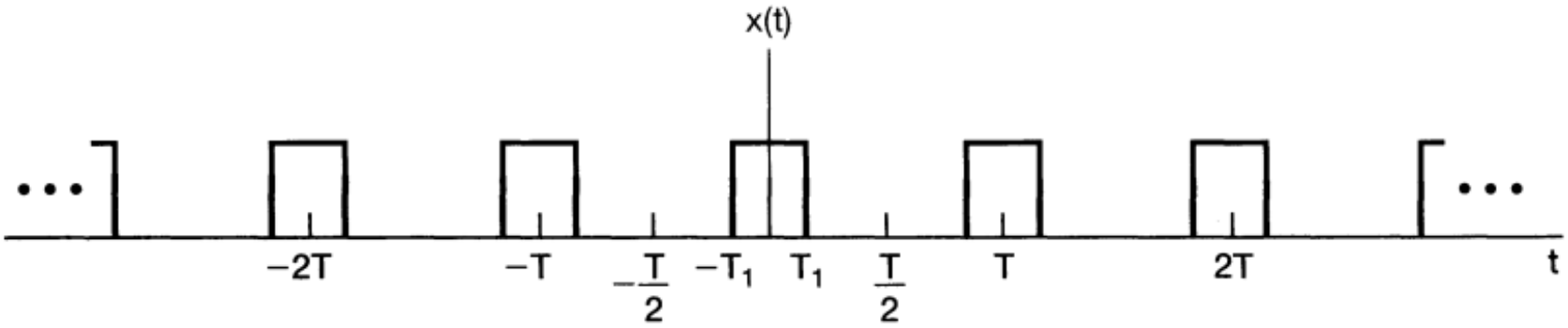


Image credit: Oppenheim chapter 4.1

Towards the Fourier transform

Initially the spacing of samples is integer multiples of $\omega = 2\pi/T$.

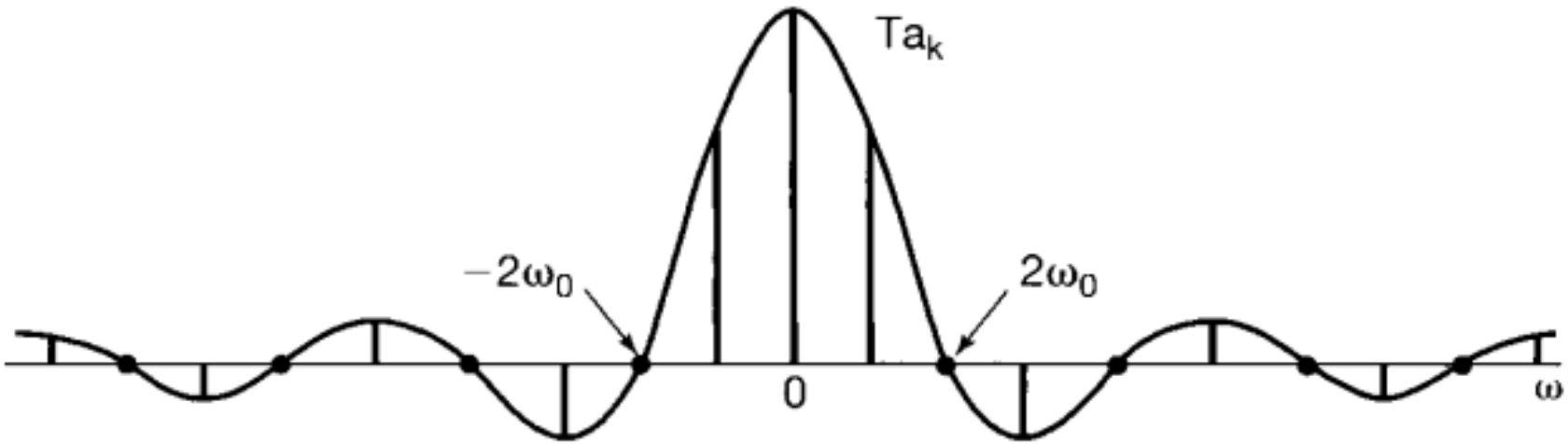
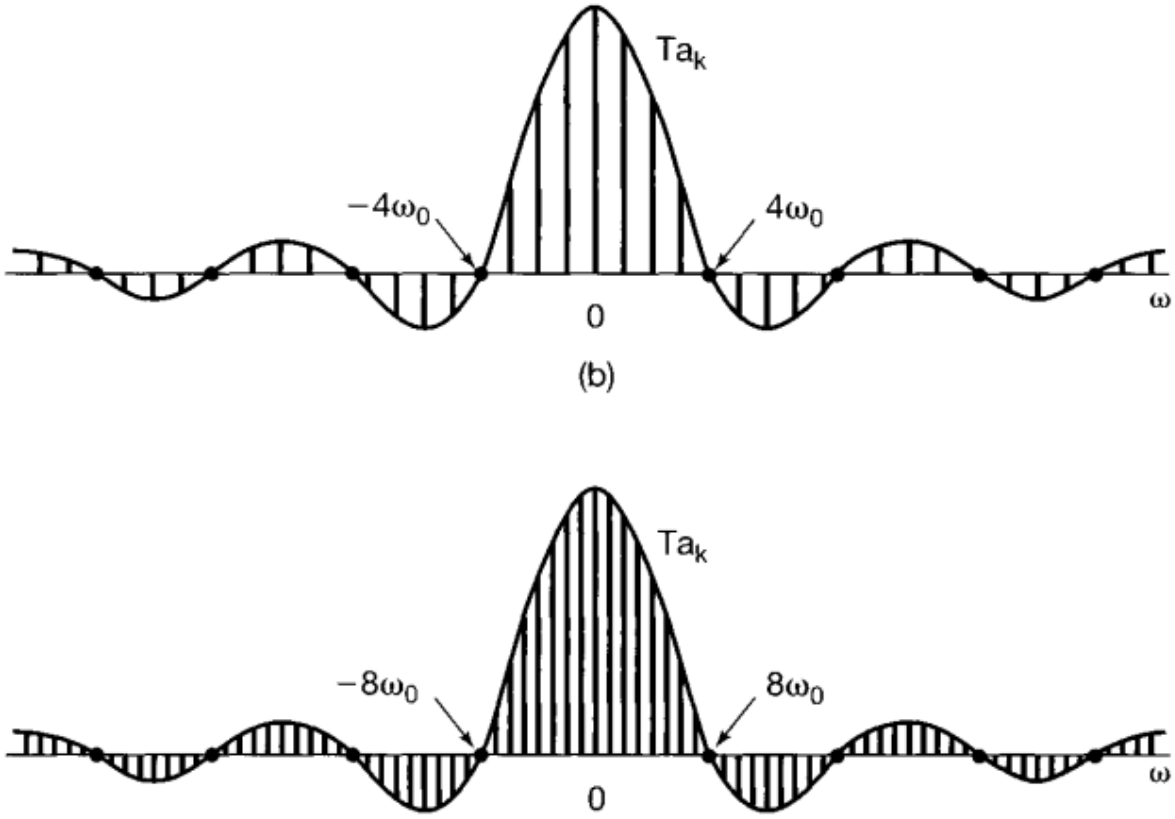


Image credit: Oppenheim chapter 4.1

Towards the Fourier transform

As T grows, $\omega = 2\pi/T$ becomes smaller so the integer multiples of it get closer together.



Towards the Fourier transform

Eventually, ω becomes so small. Instead of

consider the sum as a continuous integral over ω:

How does this apply to aperiodic signals?

Towards the Fourier transform

Given any aperiodic signal $x(t)$, "pretend" it's periodic by constructing a **periodic extension**, $\tilde{x}(t)$ with period T.

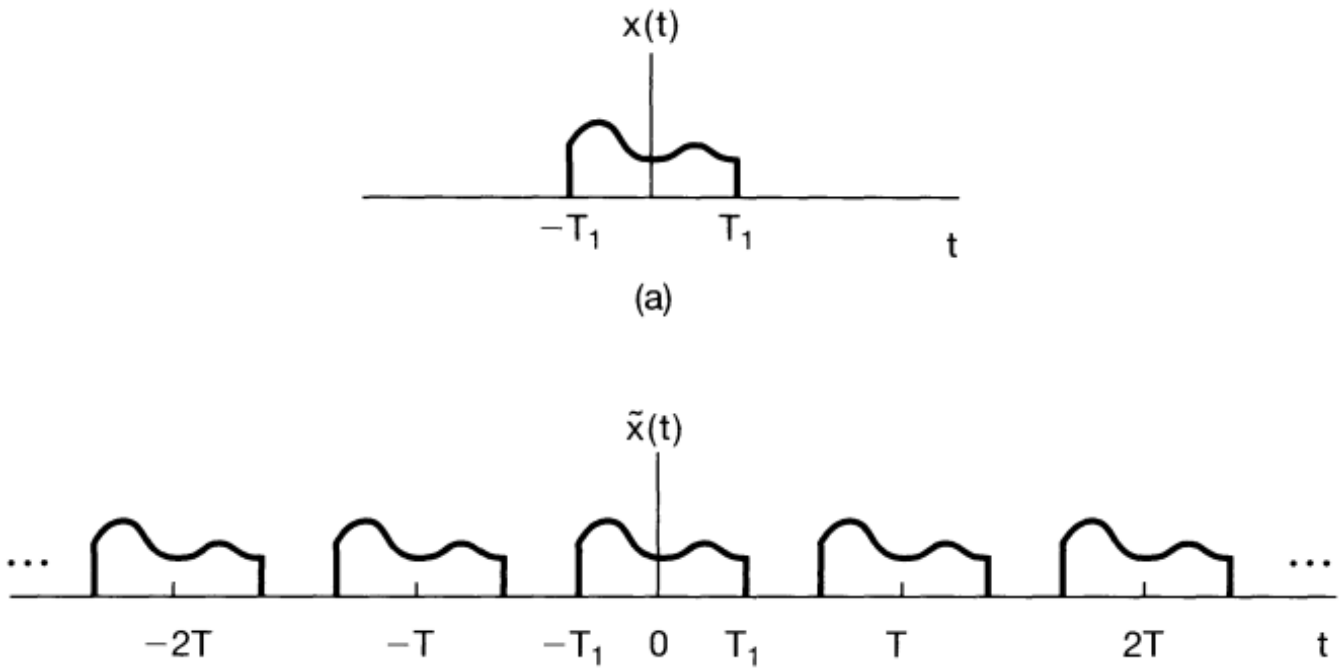


Image credit: Oppenheim chapter 4.1

Motivation: Fourier transform

We can express $\tilde{x}(t)$ as a Fourier series (where $\omega = 2\pi/T$):

Motivation: Fourier transform

What happens to the coefficients?

Let's define

so that

Motivation: Fourier transform

We can put this back in our Fourier series:

Motivation: Fourier transform

Consider what happens when $T \to \infty$...

1. $\tilde{x}(t)$ will look just like $x(t)$ for large enough T

2. ω will get smaller and smaller

The Fourier transform

Inverse Fourier transform (synthesis equation):

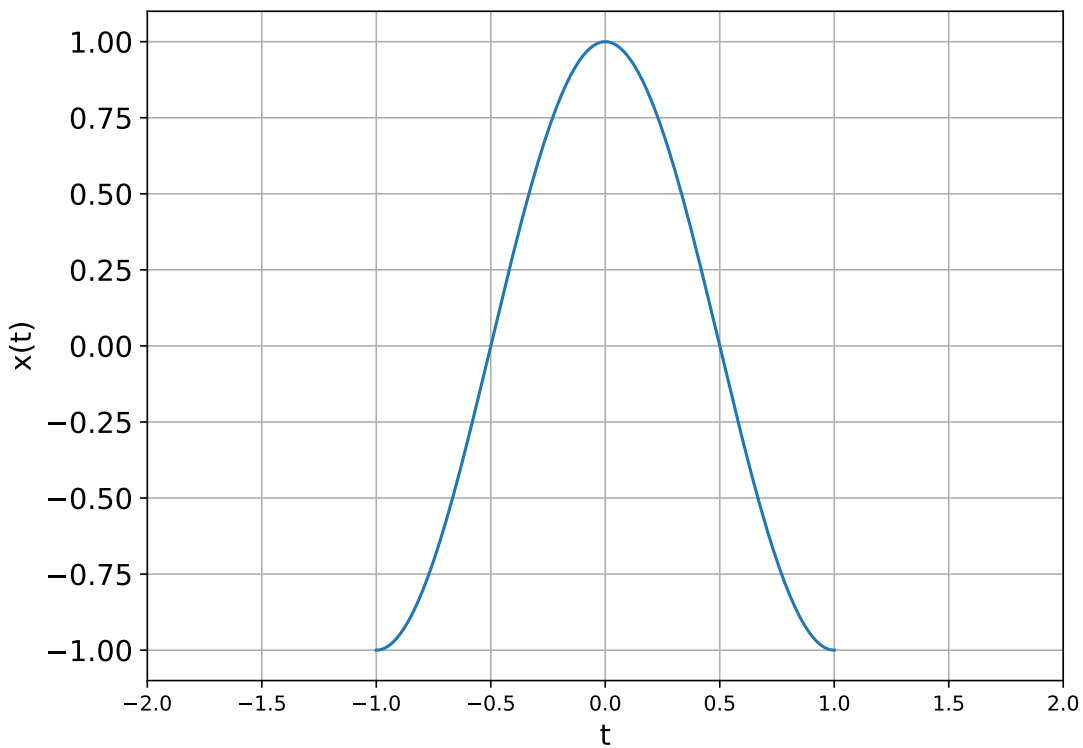
Fourier transform (analysis equation, or Fourier *spectrum*):

Note: Sometimes the $1/2\pi$ prefactor appears on the spectrum, or sometimes both versions have $1/\sqrt{2\pi}$.

Example

Compute the Fourier spectrum of:

$$x(t) = \begin{cases} \cos(\pi t), & |t| \leq 1 \\ 0, & |t| > 1 \end{cases}$$



Example

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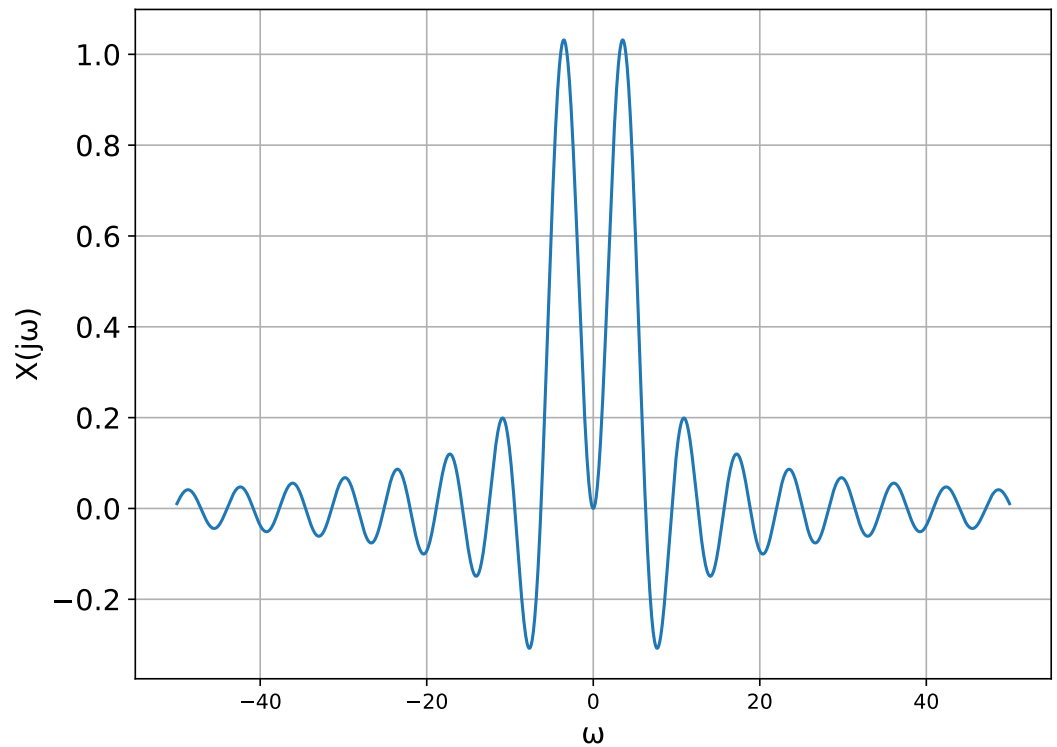
Start from the definition:

Example

$$X(j\omega) \qquad \frac{1}{2}\int_{-1}^{1} e^{j(\pi-\omega)t}dt + \frac{1}{2}\int_{-1}^{1} e^{-j(\pi+\omega)t}dt$$

Example

$$X(j\omega) = \frac{\sin(\omega)}{\pi - \omega} - \frac{\sin(\omega)}{\pi + \omega}$$



Fourier transform and impulse response

You've actually already seen the Fourier transform before...

Write a signal as a combination of shifted, weighted impulses:

Put this in an LTI system with impulse response $h(t)$:

Fourier transform and impulse response

$$x(t) \to y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$$

When the signal is a complex exponential,

Fourier transform and impulse response

The system function $H(j\omega)$, or frequency response

is the **Fourier transform of the impulse response**!

Take the inverse Fourier transform to find $h(t)$ from $H(j\omega)$:

Dirichlet conditions for Fourier transforms

If a signal

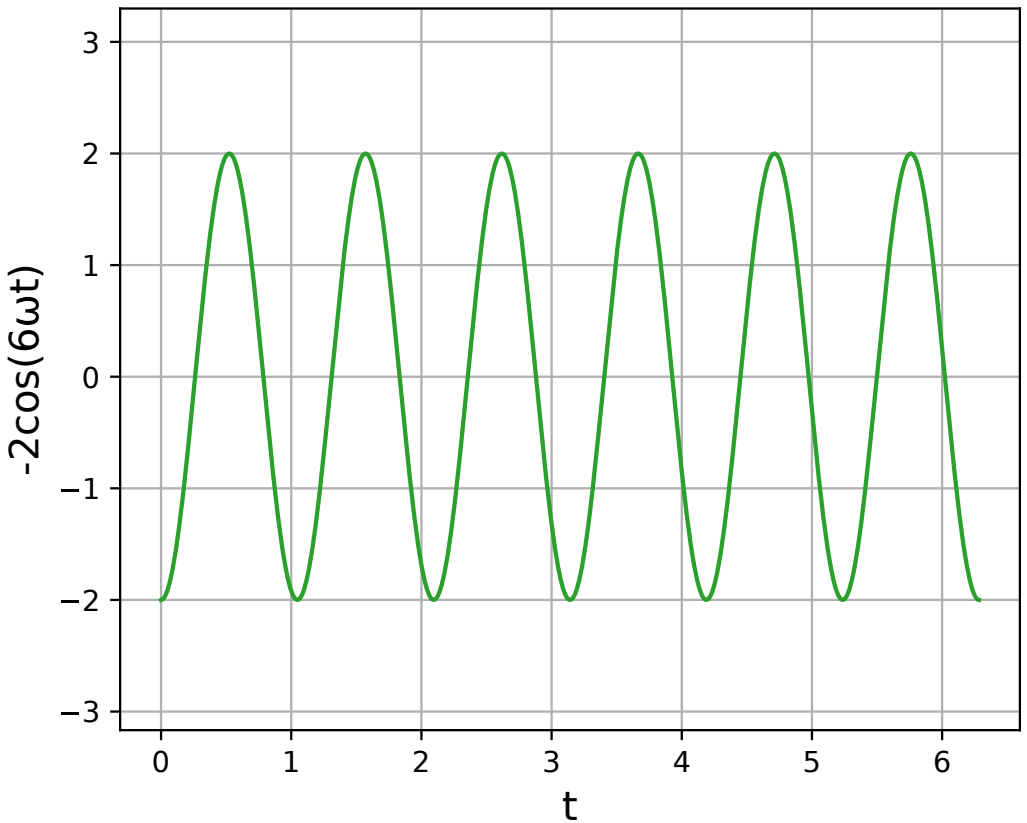
1. is single-valued
2. is absolutely integrable ($\int_{-\infty}^{\infty} |x(t)|dt < \infty$)
3. has a finite number of maxima and minima within any finite interval
4. has a finite number of finite discontinuities within any finite interval

then the Fourier transform converges to

- $x(t)$ where it is continuous
- the average of the values on either side at a discontinuity

Dirichlet conditions for Fourier transforms

The "**within any finite interval**" takes care of periodic signals:



For next time

Content:

- Dirichlet conditions for CT Fourier transform
- Convolution and multiplication property of the Fourier transform
- Time-frequency duality
- Differentiation and integration of FT

Recommended reading:

- From today's class: Oppenheim 4.0-4.2
- Suggested problems: 4.1-4.4, 4.6, 4.9, 4.21bcdgh, 4.22abde, 4.27
- For next class: Oppenheim 4.3-4.6